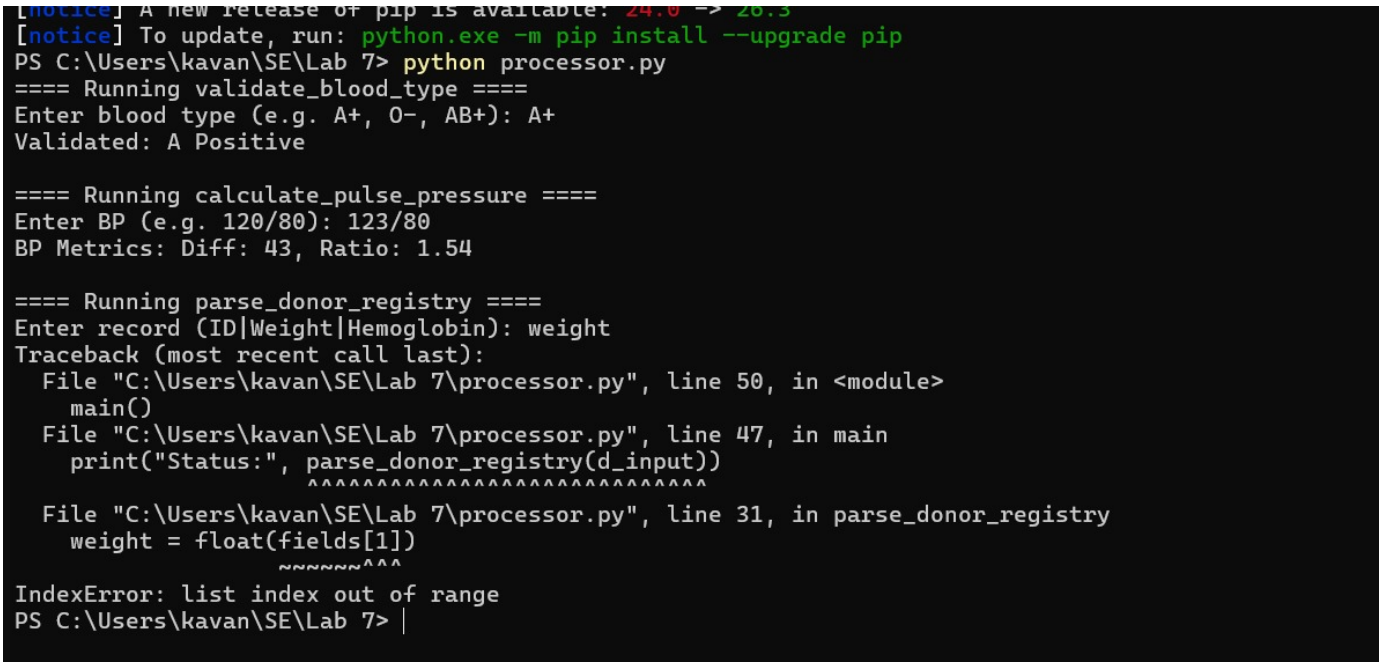
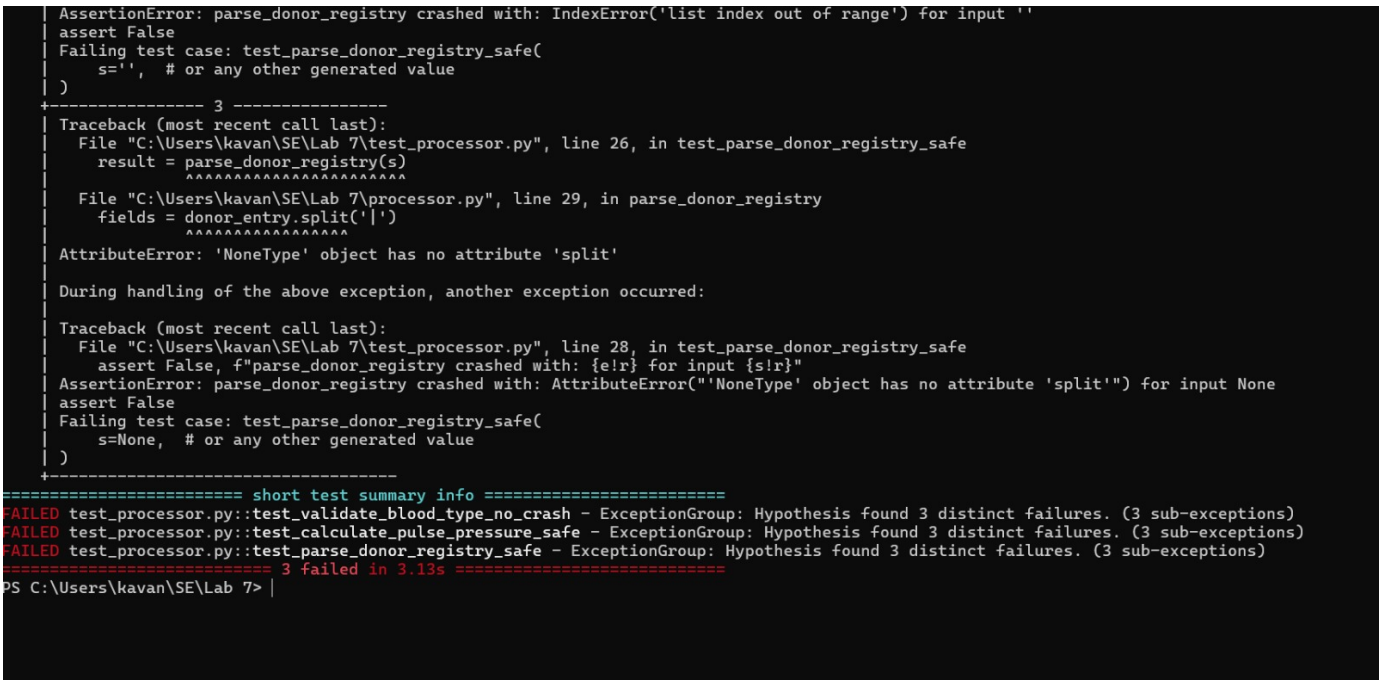
Step 2: Running the buggy processor.py

Ran python processor.py on the original code. A malformed donor record (no pipe separators) crashed parse_donor_registry with an IndexError, because the code assumed the input always had 3 fields.



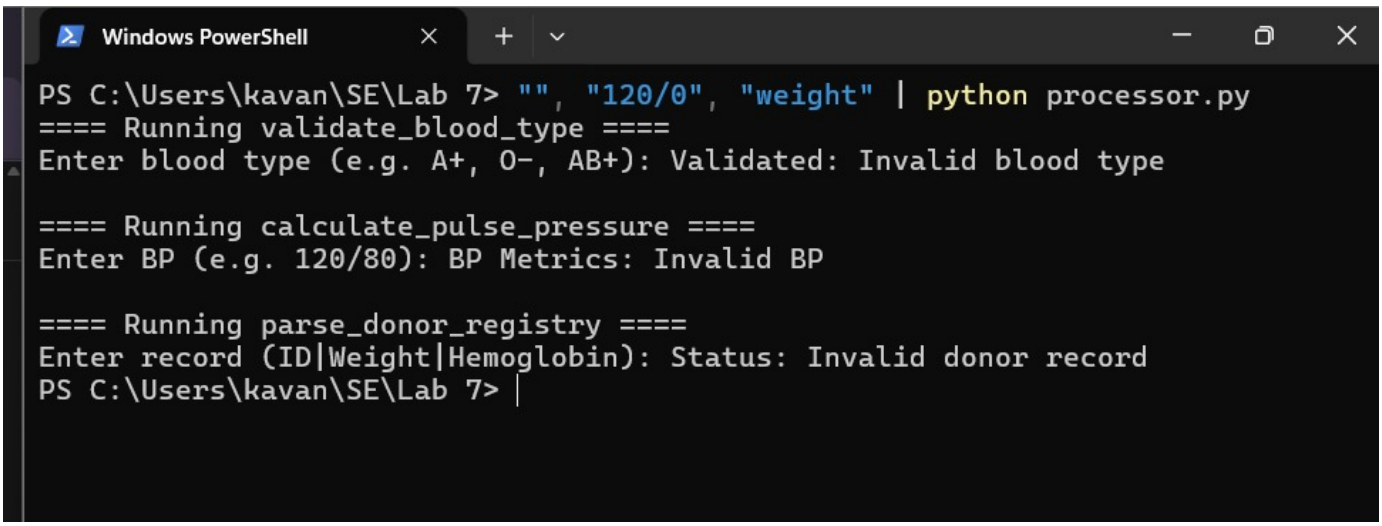
Step 4: Hypothesis tests failing on the buggy code

Ran pytest test_processor.py. All 3 fuzz tests failed. Hypothesis found crashing inputs such as an empty string (IndexError) and None (AttributeError) for all three functions.



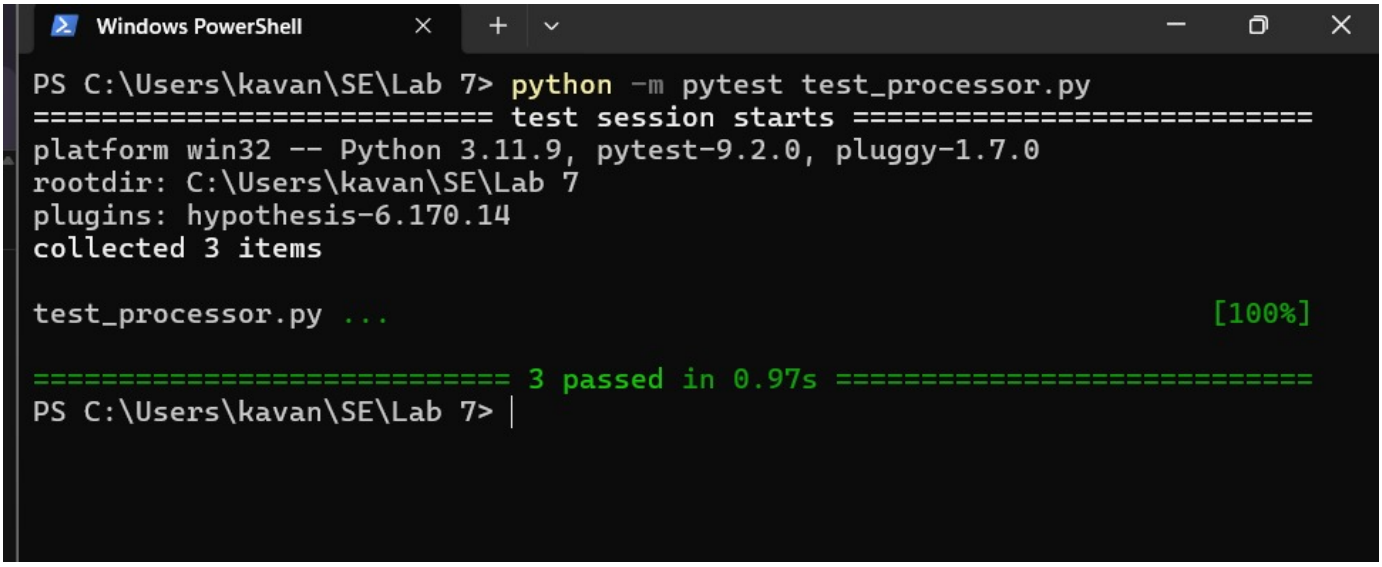
Step 5: Running the fixed processor.py

After adding input validation, the same bad inputs (empty blood type, BP 120/0, malformed donor record) now return safe Invalid messages instead of crashing.



Step 6: Hypothesis tests passing on the fixed code

Re-ran pytest test_processor.py against the fixed processor.py. All 3 tests passed.



Summary of bugs found and fixes

- validate_blood_type: crashed on None, empty or 1-character input and on invalid codes (KeyError). Fix: check the type, minimum length, valid antigen (A, B, AB, O) and Rh sign (+ or -).

- calculate_pulse_pressure: crashed on None, missing slash, non-numeric values and diastolic = 0 (ZeroDivisionError). Fix: check the type, split into exactly 2 parts, wrap int() in try/except, and require positive values.

- parse_donor_registry: crashed on None, wrong number of fields and non-numeric values. Fix: check the type, require exactly 3 fields, a non-empty ID, finite positive numbers (rejects nan and inf).

- All three functions now return a clear Invalid message instead of raising an exception.